Predicting Water Pump Functionality in Tanzania

A Machine Learning Approach to Proactive Maintenance

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59,400 records | 3 status classes | Random Forest Test F1 = 0.6677

The Problem

Act before pumps fail — not after.

- Tens of thousands of rural water points across Tanzania; many are non-functional or in active need of repair.
- Manual inspection of every pump is too slow and too expensive — crews end up reacting after failure.
- A reliable predictive model lets the Ministry of Water target inspections, plan budgets, and protect rural water access.

Stakeholder: Tanzanian Ministry of Water + rural NGOs

Data Understanding

59,400 pumps · 40 features · 3 classes

Target variable: status_group

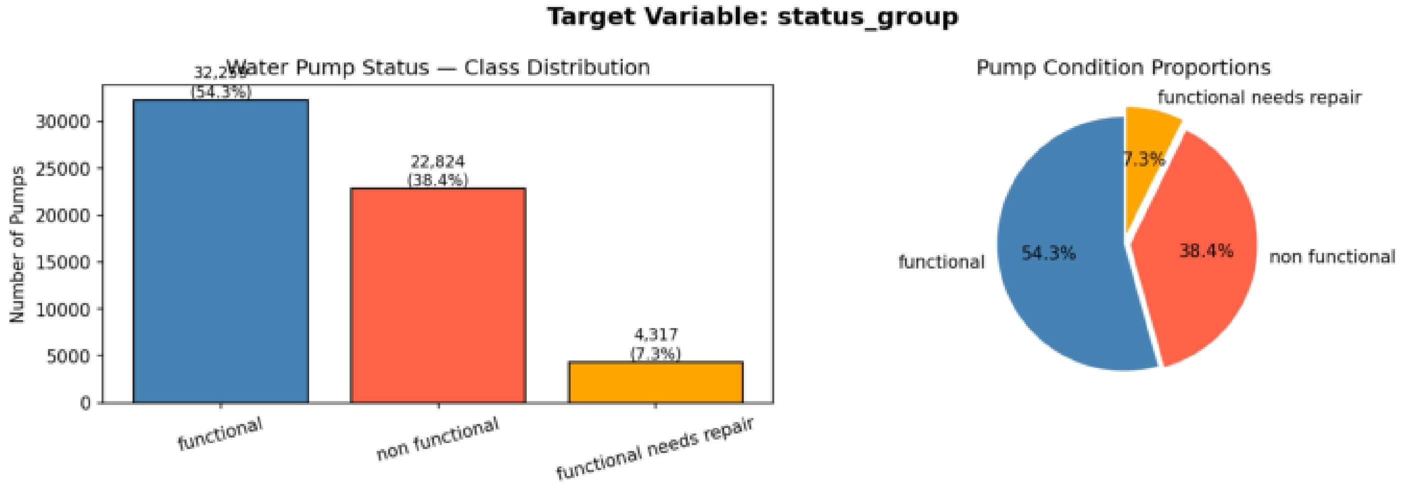
Class	Count	Share
functional	32,259	54.3%
non functional	22,824	38.4%
functional needs repair	4,317	7.3%

Imbalance matters: the smallest class is the most operationally valuable to catch — these pumps can still be saved.

Source: DrivenData "Pump It Up" — Tanzania Ministry of Water

EDA · Class Distribution

More than half functional, but ~46% need attention.

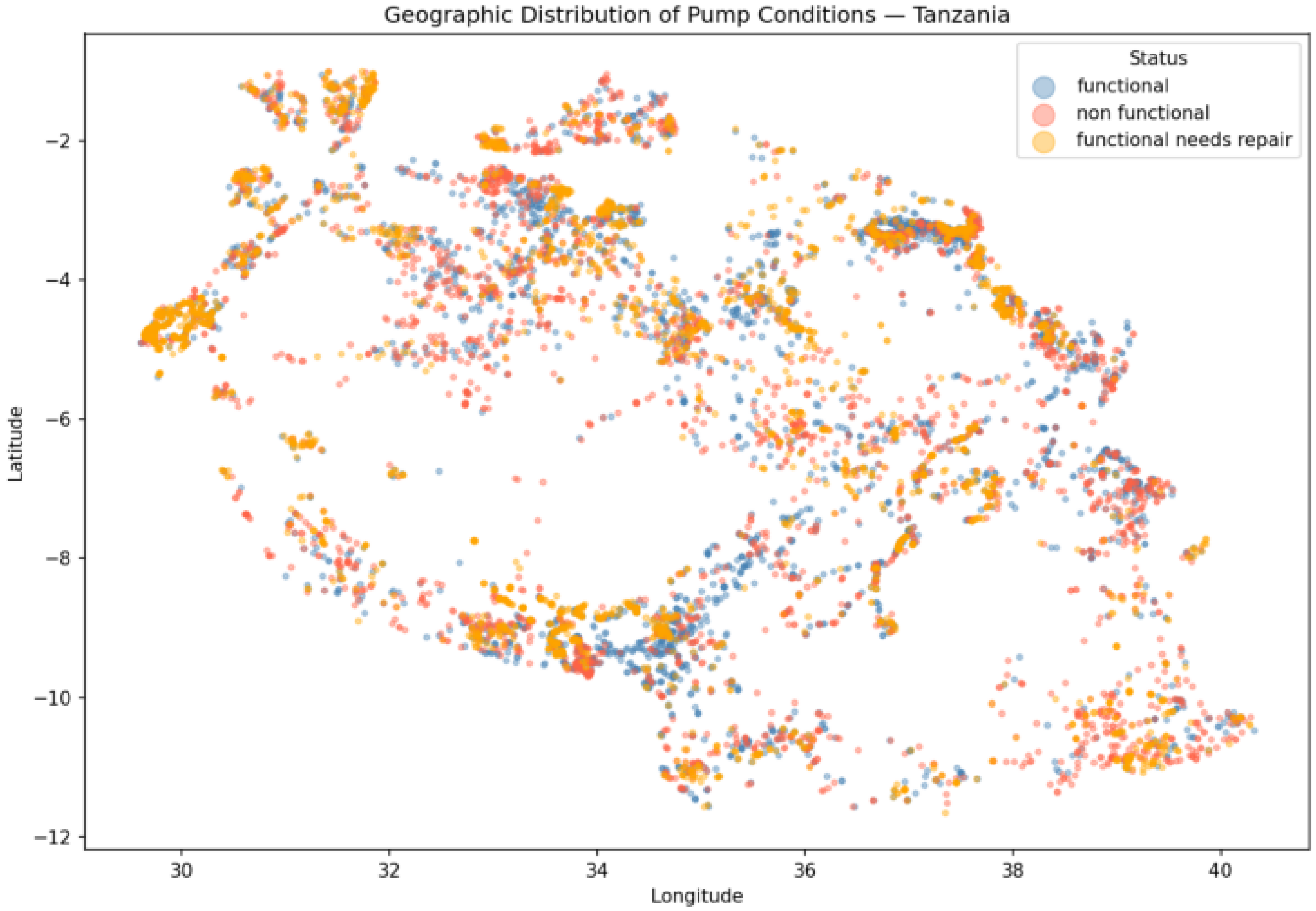


- Class imbalance: only 7.3% are "needs repair".
- Macro F1 chosen as evaluation metric.
- Naive "always functional" baseline = 54.3% accuracy but operationally useless.

Image: images\eda_01_class_distribution.png

EDA · Geographic Distribution

Failures cluster in identifiable regions.

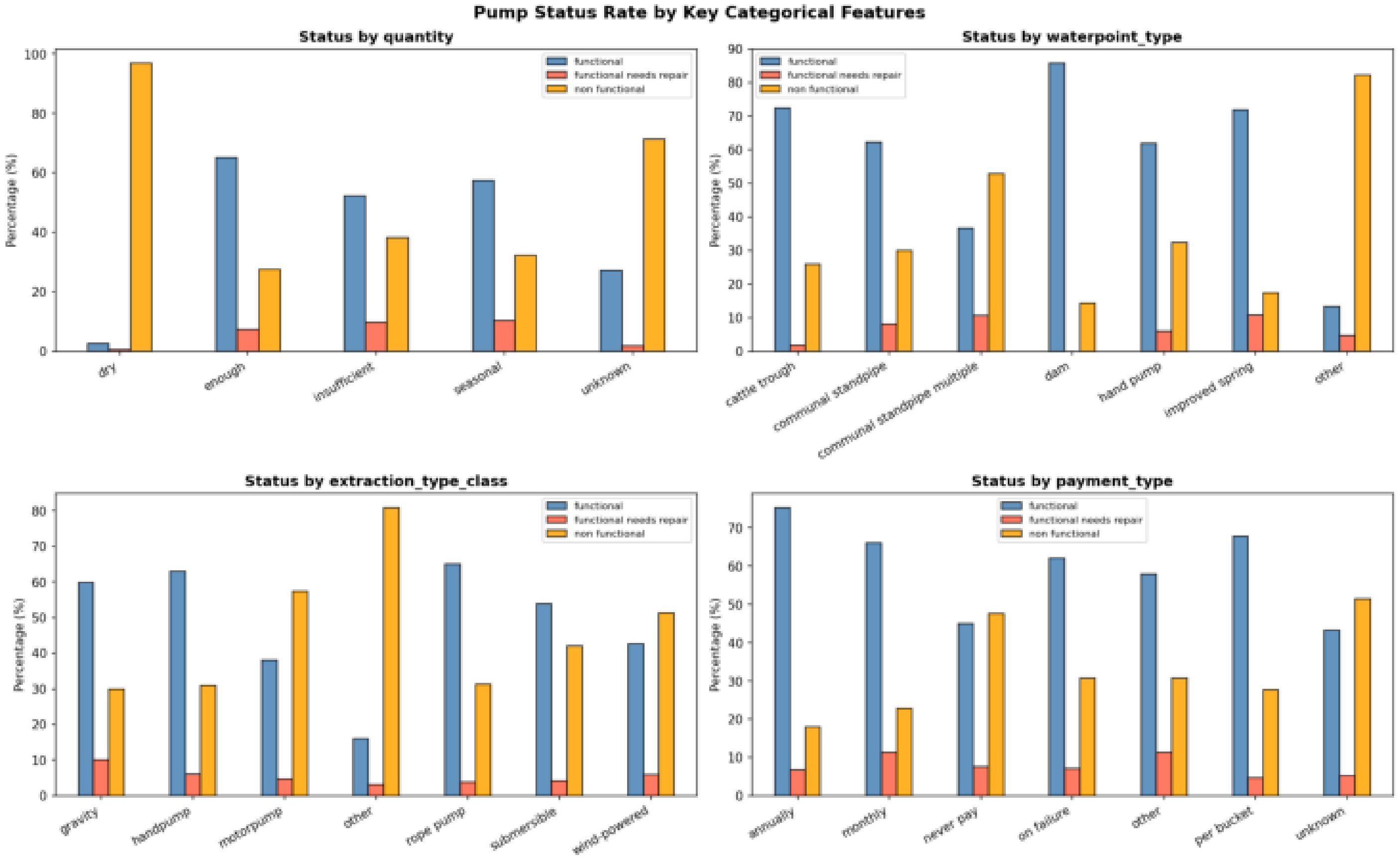


- Latitude/longitude carry real predictive signal.
- Supports regional deployment of maintenance teams.
- Reduces travel cost and response time vs. national sweeps.

Image: images\eda_04_geographic.png

EDA · Feature Patterns

Quantity, payment, and extraction type drive failure.



- "Dry" quantity reading = strongest single failure signal.
- "Never pay" waterpoints fail at notably higher rates.
- Gravity-fed extraction is more reliable than motorised.

Image: images\eda_03_categorical_vs_status.png

Modeling Comparison

Four iterations — the ensemble wins.

#	Model	Train F1	Test F1	Notes
1	Logistic Regression	~0.59	~0.58	Baseline — too linear
2	Decision Tree (default)	~0.95	~0.55	Memorised training data
3	Decision Tree (tuned)	~0.65	0.5766	Pruning fixed overfit
4	**Random Forest**	**0.7651**	**0.6677**	**FINAL MODEL**

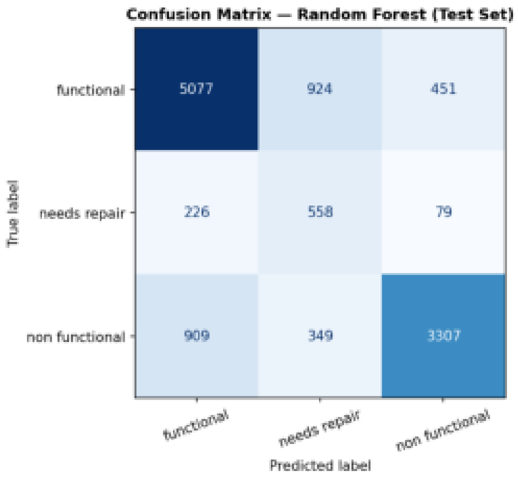
Linear models underfit; one deep tree overfits; pruning helps but caps performance.

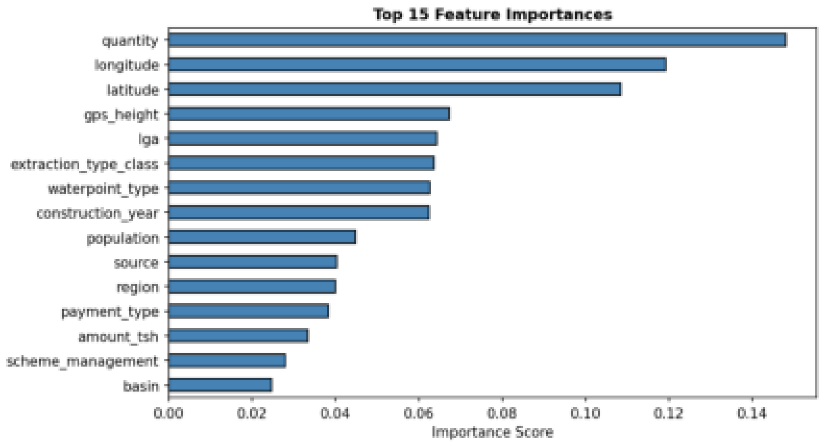
Aggregating 100 pruned trees through majority vote gives the best generalisation — like asking 100 experts to vote.

Final Evaluation

Random Forest on the holdout test set

Train F1	**0.7651**
Test F1	**0.6677**
Train/Test gap	**0.0974**





Small gap = the model generalises rather than memorises. Test performance is close to training performance.

Caveat: the minority "functional needs repair" class remains the hardest to predict — misclassifications delay maintenance until full failure.

Recommendations & Next Steps

From model output to operational action

Operational

- Triage by quantity + age — dry pumps + older builds first
- Geographic deployment — dispatch to failure-hotspot regions
- Payment scheme reform — pilot in "never pay" waterpoints
- Tablet integration — model runs at point of inspection

Technical next steps

- Cross-validated hyperparameter tuning
- Class-imbalance handling beyond class_weight (SMOTE, threshold)
- Feature engineering: pump age, regional aggregations, seasonality
- Annual retraining with fresh survey data

Tanzania’s pump failure problem is predictable from data the Ministry already collects.

Random Forest lifts hold-out F1 from 0.44 (baseline) to 0.6677 — enough to redirect inspection effort proactively.